# Spectral Embedding of Platonic Solids

In this page, we have not embedded for $\lambda_{max}$ which has multiplicity 1 for every solid. Such plots give us a single point. We are unable to add them because the file is unable to render too many interactive plots.

## Tetrahedron

```
Eigenvalues:
[-1. -1. -1.  3.]


Eigenvectors for  = -1.0:

Eigenvector 1:
[ 0.     -0.5774  0.7887 -0.2113]

Eigenvector 2:
[ 0.     -0.5774 -0.2113  0.7887]

Eigenvector 3:
[-0.866   0.2887  0.2887  0.2887]
[[ 0.          0.         -0.8660254 ]
 [-0.57735027 -0.57735027  0.28867513]
 [ 0.78867513 -0.21132487  0.28867513]
 [-0.21132487  0.78867513  0.28867513]]
```

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## Cube

```
Eigenvalues:
[-3. -1. -1. -1.  1.  1.  1.  3.]


Eigenvectors for  = -3.0:

Eigenvector 1:
```

```
[ 0.3536 -0.3536  0.3536 -0.3536 -0.3536  0.3536 -0.3536  0.3536]
[[ 0.35355339  0.          0.        ]
 [-0.35355339  0.          0.        ]
 [ 0.35355339  0.          0.        ]
 [-0.35355339  0.          0.        ]
 [-0.35355339  0.          0.        ]
 [ 0.35355339  0.          0.        ]
 [-0.35355339  0.          0.        ]
 [ 0.35355339  0.          0.        ]]
```

```
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```
Eigenvectors for  = -1.0:

Eigenvector 1:
[-0.5914  0.0852  0.3391  0.1672  0.3391  0.1672 -0.5914  0.0852]

Eigenvector 2:
[-0.1587  0.4701 -0.4761  0.1647 -0.4761  0.1647 -0.1587  0.4701]

Eigenvector 3:
[ 0.      0.3831  0.1826 -0.5656  0.1826 -0.5656  0.      0.3831]
[[-0.59144983 -0.15870444  0.        ]
 [ 0.08519785  0.47011757  0.38305456]
 [ 0.33909951 -0.47610733  0.18257419]
 [ 0.16715247  0.16469419 -0.56562874]
 [ 0.33909951 -0.47610733  0.18257419]
 [ 0.16715247  0.16469419 -0.56562874]
 [-0.59144983 -0.15870444  0.        ]
 [ 0.08519785  0.47011757  0.38305456]]
```

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```
Eigenvectors for  = 1.0000000000000002:

Eigenvector 1:
[-0.0031  0.4864  0.5127  0.0232 -0.5127 -0.0232  0.0031 -0.4864]

Eigenvector 2:
[-0.6124 -0.2064  0.2013 -0.2047 -0.2013  0.2047  0.6124  0.2064]

Eigenvector 3:
[ 5.000e-04  3.096e-01 -2.675e-01 -5.767e-01  2.675e-01  5.767e-
01
 -5.000e-04 -3.096e-01]
[[-3.13401981e-03 -6.12364215e-01  4.95824266e-04]
```

```
 [ 4.86358812e-01 -2.06365321e-01  3.09626324e-01]
 [ 5.12742375e-01  2.01286094e-01 -2.67542830e-01]
 [ 2.32495428e-02 -2.04712800e-01 -5.76673329e-01]
 [-5.12742375e-01 -2.01286094e-01  2.67542830e-01]
 [-2.32495428e-02  2.04712800e-01  5.76673329e-01]
 [ 3.13401981e-03  6.12364215e-01 -4.95824266e-04]
 [-4.86358812e-01  2.06365321e-01 -3.09626324e-01]]
```

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### Octahedron

```
Eigenvalues:
[-2. -2. -0. -0.  0.  4.]


Eigenvectors for  = -2.0000000000000004:

Eigenvector 1:
[ 4.000e-04 -5.002e-01  4.998e-01 -5.002e-01  4.998e-01  4.000e-
04]

Eigenvector 2:
[ 0.5774 -0.2883 -0.2891 -0.2883 -0.2891  0.5774]
[[ 4.44410839e-04  5.77350098e-01  0.00000000e+00]
 [-5.00222057e-01 -2.88290178e-01  0.00000000e+00]
 [ 4.99777646e-01 -2.89059920e-01  0.00000000e+00]
 [-5.00222057e-01 -2.88290178e-01  0.00000000e+00]
 [ 4.99777646e-01 -2.89059920e-01  0.00000000e+00]
 [ 4.44410839e-04  5.77350098e-01  0.00000000e+00]]
```

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```
Eigenvectors for  = -2.993751573463213e-17:

Eigenvector 1:
[-0.6972  0.1163 -0.0195 -0.1163  0.0195  0.6972]

Eigenvector 2:
[-0.0365 -0.1014  0.6988  0.1014 -0.6988  0.0365]

Eigenvector 3:
[-0.1121 -0.6901 -0.106   0.6901  0.106   0.1121]
[[-0.69720813 -0.03648073 -0.1121159 ]
 [ 0.1162742  -0.10141064 -0.6900697 ]
 [-0.01952251  0.6988455  -0.10598977]
 [-0.1162742   0.10141064  0.6900697 ]
```

```
 [ 0.01952251 -0.6988455   0.10598977]
 [ 0.69720813  0.03648073  0.1121159 ]]
```

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## Dodecahedron

```
Eigenvalues:
[-2.2361 -2.2361 -2.2361 -2.     -2.     -2.     -2.     -0.      0.
  0.      0.      1.      1.      1.      1.      1.      2.2361  2.2361
  2.2361  3.    ]


Eigenvectors for  = -2.2360679774997894:

Eigenvector 1:
[-0.3873  0.2887 -0.1291 -0.1291  0.2887  0.2887 -0.1291 -0.1291  0.1291
  0.1291 -0.2887  0.1291  0.1291 -0.1291 -0.1291  0.1291  0.1291 -
0.2887
  0.3873 -0.2887]

Eigenvector 2:
[-0.      0.2528 -0.2373  0.3029 -0.081  -0.1718  0.359  -0.328   0.1217
 -0.0251  0.1718 -0.359   0.328  -0.1217  0.0251  0.2373 -0.3029  0.081
  0.     -0.2528]

Eigenvector 3:
[ 0.     -0.0524  0.2775 -0.2039  0.2451 -0.1928  0.0668 -0.1604  0.3443
 -0.3643  0.1928 -0.0668  0.1604 -0.3443  0.3643 -0.2775  0.2039 -
0.2451
 -0.      0.0524]
[[-3.87298335e-01 -6.55145067e-18  2.32990222e-36]
 [ 2.88675135e-01  2.52829231e-01 -5.23836466e-02]
 [-1.29099445e-01 -2.37306105e-01  2.77523235e-01]
 [-1.29099445e-01  3.02920297e-01 -2.03893666e-01]
 [ 2.88675135e-01 -8.10490470e-02  2.45148360e-01]
 [ 2.88675135e-01 -1.71780184e-01 -1.92764714e-01]
 [-1.29099445e-01  3.58995224e-01  6.67514984e-02]
 [-1.29099445e-01 -3.28037243e-01 -1.60389840e-01]
 [ 1.29099445e-01  1.21689119e-01  3.44274733e-01]
 [ 1.29099445e-01 -2.51169456e-02 -3.64283505e-01]
 [-2.88675135e-01  1.71780184e-01  1.92764714e-01]
 [ 1.29099445e-01 -3.58995224e-01 -6.67514984e-02]
 [ 1.29099445e-01  3.28037243e-01  1.60389840e-01]
 [-1.29099445e-01 -1.21689119e-01 -3.44274733e-01]
 [-1.29099445e-01  2.51169456e-02  3.64283505e-01]
 [ 1.29099445e-01  2.37306105e-01 -2.77523235e-01]
```

```
 [ 1.29099445e-01 -3.02920297e-01  2.03893666e-01]
 [-2.88675135e-01  8.10490470e-02 -2.45148360e-01]
 [ 3.87298335e-01  5.55111512e-17 -2.77555756e-17]
 [-2.88675135e-01 -2.52829231e-01  5.23836466e-02]]
```

```
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```
Eigenvectors for  = -2.0000000000000004:

Eigenvector 1:
[-0.4472  0.2981 -0.0745 -0.0745  0.2981  0.2981 -0.0745 -0.0745 -
0.0745
 -0.0745  0.2981 -0.0745 -0.0745 -0.0745 -0.0745 -0.0745 -0.0745  0.2981
 -0.4472  0.2981]

Eigenvector 2:
[ 0.      0.3316 -0.3318  0.1363 -0.136  -0.1956  0.1953 -0.3313  0.1357
  0.1958 -0.1956  0.1953 -0.3313  0.1357  0.1958 -0.3318  0.1363 -
0.136
 -0.      0.3316]

Eigenvector 3:
[-0.      0.0338 -0.0855  0.3511 -0.2994  0.2656 -0.3173  0.018   0.2476
 -0.2138  0.2656 -0.3173  0.018   0.2476 -0.2138 -0.0855  0.3511 -
0.2994
  0.      0.0338]

Eigenvector 4:
[-0.      0.0065  0.2775 -0.2293 -0.0547  0.0482  0.2358 -0.2905  0.3387
 -0.3322  0.0482  0.2358 -0.2905  0.3387 -0.3322  0.2775 -0.2293 -
0.0547
  0.      0.0065]

Applying PCA (dim > 3 → reducing to 3D)
[[-4.45937567e-18 -1.02620246e-18  1.26300820e-17]
 [-2.97898190e-01  7.76657484e-02 -1.27811622e-01]
 [ 1.83059541e-01 -3.10541013e-01  2.53964423e-01]
 [ 2.57887955e-02  1.14932924e-01 -4.24935060e-01]
 [ 8.90498538e-02  1.17942341e-01  2.98782260e-01]
 [ 2.08848337e-01 -1.95608089e-01 -1.70970638e-01]
 [-3.23686986e-01 -3.72671756e-02  2.97123438e-01]
 [ 4.12736840e-01  1.55209516e-01  1.65882221e-03]
 [-2.03888503e-01 -3.50817606e-01 -1.72629460e-01]
 [-9.40096872e-02  4.28483354e-01  4.48178372e-02]
 [ 2.08848337e-01 -1.95608089e-01 -1.70970638e-01]
 [-3.23686986e-01 -3.72671756e-02  2.97123438e-01]
```

```
 [ 4.12736840e-01  1.55209516e-01  1.65882221e-03]
 [-2.03888503e-01 -3.50817606e-01 -1.72629460e-01]
 [-9.40096872e-02  4.28483354e-01  4.48178372e-02]
 [ 1.83059541e-01 -3.10541013e-01  2.53964423e-01]
 [ 2.57887955e-02  1.14932924e-01 -4.24935060e-01]
 [ 8.90498538e-02  1.17942341e-01  2.98782260e-01]
 [ 1.08250435e-16 -2.36079120e-16 -1.86692695e-16]
 [-2.97898190e-01  7.76657484e-02 -1.27811622e-01]]
```

```
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```
Eigenvectors for   = -5.5161000268398e-16:

Eigenvector 1:
[ 2.000e-04  3.383e-01  2.532e-01 -2.552e-01 -3.364e-01 -1.800e-
03
 -8.320e-02 -2.534e-01 -2.551e-01 -8.300e-02  1.800e-03  8.320e-
02
  2.534e-01  2.551e-01  8.300e-02 -2.532e-01  2.552e-01  3.364e-
01
 -2.000e-04 -3.383e-01]

Eigenvector 2:
[ 0.4285  0.0526 -0.2679 -0.2866  0.0734 -0.126  -0.1945 -0.1606  0.1419
  0.234   0.126   0.1945  0.1606 -0.1419 -0.234   0.2679  0.2866 -
0.0734
 -0.4285 -0.0526]

Eigenvector 3:
[ 0.1222 -0.2364  0.1773  0.1268 -0.1898  0.4263 -0.0126 -0.2995  0.249
  0.1097 -0.4263  0.0126  0.2995 -0.249  -0.1097 -0.1773 -0.1268  0.1898
 -0.1222  0.2364]

Eigenvector 4:
[-0.0384 -0.164   0.1808 -0.1914  0.213  -0.049   0.3938 -0.1424 -
0.2298
  0.3554  0.049  -0.3938  0.1424  0.2298 -0.3554 -0.1808  0.1914 -
0.213
  0.0384  0.164 ]

Applying PCA (dim > 3 → reducing to 3D)
[[ 0.4291223   0.11513115  0.05047437]
 [ 0.0190251  -0.01595855 -0.18308664]
 [-0.25885554 -0.05588081  0.32488318]
 [-0.28255537  0.22860406 -0.06462994]
 [ 0.09326351 -0.27189585 -0.12764096]
```

```
 [-0.11228861  0.2878544   0.3107276 ]
 [-0.16559203 -0.32777666  0.19724221]
 [-0.17026676 -0.05925034 -0.37535754]
 [ 0.14656693  0.34373521 -0.01415557]
 [ 0.26353027 -0.21264551  0.24771658]
 [ 0.11228861 -0.2878544  -0.3107276 ]
 [ 0.16559203  0.32777666 -0.19724221]
 [ 0.17026676  0.05925034  0.37535754]
 [-0.14656693 -0.34373521  0.01415557]
 [-0.26353027  0.21264551 -0.24771658]
 [ 0.25885554  0.05588081 -0.32488318]
 [ 0.28255537 -0.22860406  0.06462994]
 [-0.09326351  0.27189585  0.12764096]
 [-0.4291223  -0.11513115 -0.05047437]
 [-0.0190251   0.01595855  0.18308664]]
```

```
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```
Eigenvectors for  = 1.0000000000000002:

Eigenvector 1:
[ 0.0005 -0.4371 -0.0814  0.2351  0.2762  0.1614  0.0403 -0.3562  0.0406
  0.1206  0.1614  0.0403 -0.3562  0.0406  0.1206 -0.0814  0.2351  0.2762
  0.0005 -0.4371]

Eigenvector 2:
[-0.0171 -0.0122  0.166  -0.2511 -0.3065  0.3017 -0.1106 -0.1611 -
0.0383
  0.4293  0.3017 -0.1106 -0.1611 -0.0383  0.4293  0.166  -0.2511 -
0.3065
 -0.0171 -0.0122]

Eigenvector 3:
[-0.4984 -0.1536  0.1923  0.1948 -0.1535 -0.1913  0.1561  0.1526  0.15
  0.151  -0.1913  0.1561  0.1526  0.15    0.151   0.1923  0.1948 -
0.1535
 -0.4984 -0.1536]

Eigenvector 4:
[-0.0361 -0.1846 -0.4183 -0.275  -0.0117  0.1601  0.1549  0.2699  0.2995
  0.0413  0.1601  0.1549  0.2699  0.2995  0.0413 -0.4183 -0.275  -
0.0117
 -0.0361 -0.1846]

Eigenvector 5:
[ 0.0043  0.0334  0.0621 -0.1345  0.2368 -0.2659 -0.4333 -0.0329  0.367
```

```
  0.1631 -0.2659 -0.4333 -0.0329  0.367   0.1631  0.0621 -0.1345  0.2368
  0.0043  0.0334]

Applying PCA (dim > 3 → reducing to 3D)
[[-0.01309071 -0.02197957 -0.49073227]
 [-0.06321089  0.35946546 -0.13922547]
 [-0.21002275  0.30172833  0.11389652]
 [-0.11858759  0.1637438   0.11776621]
 [-0.30700925 -0.20403105 -0.14368668]
 [ 0.35712943 -0.17741397 -0.20782011]
 [ 0.39844441  0.06604652  0.14755637]
 [ 0.15990257  0.0797167   0.23761027]
 [-0.17533095 -0.34579529  0.22927938]
 [-0.02822427 -0.22148093  0.13535579]
 [ 0.35712943 -0.17741397 -0.20782011]
 [ 0.39844441  0.06604652  0.14755637]
 [ 0.15990257  0.0797167   0.23761027]
 [-0.17533095 -0.34579529  0.22927938]
 [-0.02822427 -0.22148093  0.13535579]
 [-0.21002275  0.30172833  0.11389652]
 [-0.11858759  0.1637438   0.11776621]
 [-0.30700925 -0.20403105 -0.14368668]
 [-0.01309071 -0.02197957 -0.49073227]
 [-0.06321089  0.35946546 -0.13922547]]
```

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```
Eigenvectors for  = 2.2360679774997902:

Eigenvector 1:
[ 0.0012  0.2585  0.3036  0.0741 -0.1128 -0.1431  0.0251  0.2732  0.3275
  0.3462  0.1431 -0.0251 -0.2732 -0.3275 -0.3462 -0.3036 -0.0741  0.1128
 -0.0012 -0.2585]

Eigenvector 2:
[-0.3863 -0.2884 -0.1136 -0.1035 -0.2721 -0.3034 -0.1542 -0.145   0.1185
  0.1379  0.3034  0.1542  0.145  -0.1185 -0.1379  0.1136  0.1035  0.2721
  0.3863  0.2884]

Eigenvector 3:
[ 0.0271  0.0027  0.212   0.3658  0.2515 -0.1936 -0.3544 -0.2331 -
0.1695
  0.1056  0.1936  0.3544  0.2331  0.1695 -0.1056 -0.212  -0.3658 -
0.2515
 -0.0271 -0.0027]
[[ 0.00116566 -0.38634426  0.02714321]
```

```
 [ 0.25847264 -0.28841725  0.00271737]
 [ 0.30355606 -0.11360749  0.21200721]
 [ 0.07411217 -0.10349612  0.36578127]
 [-0.11277537 -0.27205672  0.25152904]
 [-0.14309076 -0.30341806 -0.19355235]
 [ 0.02506084 -0.15423984 -0.35437555]
 [ 0.27324067 -0.14496883 -0.23307418]
 [ 0.32745124  0.11849694 -0.16951154]
 [ 0.34618718  0.13787931  0.10556388]
 [ 0.14309076  0.30341806  0.19355235]
 [-0.02506084  0.15423984  0.35437555]
 [-0.27324067  0.14496883  0.23307418]
 [-0.32745124 -0.11849694  0.16951154]
 [-0.34618718 -0.13787931 -0.10556388]
 [-0.30355606  0.11360749 -0.21200721]
 [-0.07411217  0.10349612 -0.36578127]
 [ 0.11277537  0.27205672 -0.25152904]
 [-0.00116566  0.38634426 -0.02714321]
 [-0.25847264  0.28841725 -0.00271737]]
```

```
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## Icosahedron

```
Eigenvalues:
[-2.2361 -2.2361 -2.2361 -1.     -1.     -1.     -1.     -1.      2.2361
  2.2361  2.2361  5.    ]


Eigenvectors for  = -2.2360679774997907:

Eigenvector 1:
[ 0.4999 -0.2331 -0.2148 -0.2248 -0.2169  0.2282  0.2248  0.2169 -
0.2282
  0.2331  0.2148 -0.4999]

Eigenvector 2:
[-2.000e-04  7.420e-02  1.994e-01  4.424e-01 -3.191e-01  3.964e-
01
 -4.424e-01  3.191e-01 -3.964e-01 -7.420e-02 -1.994e-01  2.000e-
04]

Eigenvector 3:
[ 0.0109  0.4361 -0.4051  0.0609 -0.318  -0.2018 -0.0609  0.318   0.2018
 -0.4361  0.4051 -0.0109]
[[ 4.99881784e-01 -2.27616573e-04  1.08697094e-02]
 [-2.33107981e-01  7.42027353e-02  4.36067224e-01]
```

```
 [-2.14759687e-01  1.99384833e-01 -4.05122161e-01]
 [-2.24783325e-01  4.42446368e-01  6.09398729e-02]
 [-2.16889394e-01 -3.19079090e-01 -3.18036988e-01]
 [ 2.28229263e-01  3.96445881e-01 -2.01846643e-01]
 [ 2.24783325e-01 -4.42446368e-01 -6.09398729e-02]
 [ 2.16889394e-01  3.19079090e-01  3.18036988e-01]
 [-2.28229263e-01 -3.96445881e-01  2.01846643e-01]
 [ 2.33107981e-01 -7.42027353e-02 -4.36067224e-01]
 [ 2.14759687e-01 -1.99384833e-01  4.05122161e-01]
 [-4.99881784e-01  2.27616573e-04 -1.08697094e-02]]
```

```
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```
Eigenvectors for  = -1.0000000000000002:

Eigenvector 1:
[ 2.000e-04 -2.919e-01 -4.691e-01  2.831e-01  2.238e-01  2.539e-
01
  2.831e-01  2.238e-01  2.539e-01 -2.919e-01 -4.691e-01  2.000e-
04]

Eigenvector 2:
[ 0.1028 -0.5464  0.3768 -0.0623 -0.0708  0.1999 -0.0623 -0.0708  0.1999
 -0.5464  0.3768  0.1028]

Eigenvector 3:
[-0.6326  0.0294  0.1955  0.0871  0.1894  0.1312  0.0871  0.1894  0.1312
  0.0294  0.1955 -0.6326]

Eigenvector 4:
[ 0.0767 -0.1094  0.0201 -0.2927  0.5707 -0.2654 -0.2927  0.5707 -
0.2654
 -0.1094  0.0201  0.0767]

Eigenvector 5:
[-0.0012  0.1417 -0.1263 -0.4893  0.0013  0.4739 -0.4893  0.0013  0.4739
  0.1417 -0.1263 -0.0012]

Applying PCA (dim > 3 → reducing to 3D)
[[ 0.00581536  0.0292582   0.02353414]
 [-0.43561786  0.11811885  0.19572898]
 [ 0.3316647  -0.46011088  0.25493581]
 [-0.08808949  0.15849584  0.26744772]
 [-0.20277631 -0.26673727 -0.52829695]
 [ 0.38900359  0.42097526 -0.2133497 ]
 [-0.08808949  0.15849584  0.26744772]
```

```
 [-0.20277631 -0.26673727 -0.52829695]
 [ 0.38900359  0.42097526 -0.2133497 ]
 [-0.43561786  0.11811885  0.19572898]
 [ 0.3316647  -0.46011088  0.25493581]
 [ 0.00581536  0.0292582   0.02353414]]
```

```
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```
Eigenvectors for   = 2.236067977499791:

Eigenvector 1:
[ 0.0159 -0.2173  0.3055 -0.0387 -0.4299 -0.4158  0.0387  0.4299  0.4158
  0.2173 -0.3055 -0.0159]

Eigenvector 2:
[-0.0321  0.3712  0.3186 -0.4584 -0.109   0.1942  0.4584  0.109  -
0.1942
 -0.3712 -0.3186  0.0321]

Eigenvector 3:
[-0.4987 -0.255  -0.235  -0.1959 -0.2308  0.1985  0.1959  0.2308 -
0.1985
  0.255   0.235   0.4987]
[[ 0.0158513  -0.0321258  -0.49871502]
 [-0.21725541  0.37118156 -0.25499871]
 [ 0.30545229  0.31855361 -0.23499468]
 [-0.03865737 -0.45840558 -0.19588245]
 [-0.42992688 -0.10900563 -0.23082602]
 [-0.41583195  0.19415944  0.19845882]
 [ 0.03865737  0.45840558  0.19588245]
 [ 0.42992688  0.10900563  0.23082602]
 [ 0.41583195 -0.19415944 -0.19845882]
 [ 0.21725541 -0.37118156  0.25499871]
 [-0.30545229 -0.31855361  0.23499468]
 [-0.0158513   0.0321258   0.49871502]]
```

```
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